

A. Questions using if (1–4)

1. Write an R program to read the dataset and print **"Century Scored"** if the first player's **Score** is greater than or equal to 100.

```
data <- read.csv("D:/Virat.csv")
if(data$Score[1] >= 100){
  print("Century Scored")
}
```

```
> data <- read.csv("D:/Virat.csv")
> if(data$Score[1] >= 100){
+   print("Century Scored")
+ }
[1] "Century Scored"
> |
```

2. Write an R program to check whether the first match was played in **India (H)**. If yes, display **"Home Match"**.

```
if(data$H.A[1] == "H"){
  print("Home Match")
}else{
  print("Not Home Match")
}

> if(data$H.A[1] == "H"){
+   print("Home Match")
+ }else{
+   print("Not Home Match")
+ }
[1] "Not Home Match"
> |
```

3. Write an R program to check whether the first match result is **Won**. If true, print **"Winning Match"**.

```
if(data$Result[1] == "Won"){
  print("Winning Match")
}else{
  print("Not Winning Match")
}
```

```

> if(data$Result[1] == "Won"){
+   print("Winning Match")
+ }else{
+   print("Not Winning Match")
+ }
[1] "Not Winning Match"
> |

```

4. Write an R program to check whether the first match was played in **Test** format. If true, print **"Test Match Played"**.

```

if(data$Format[1] == "Test"){
  print("Test Match Played")
}

> if(data$Format[1] == "Test"){
+   print("Test Match Played")
+ }
[1] "Test Match Played"
> |

```

B. Questions using if...else (5–8)

5. Write an R program to classify the first score:
- Score $\geq 100 \rightarrow$ "Century"
 - Otherwise $\rightarrow$ "Less than Century"

```

if(data$Score[1] >= 100){
  print("Century")
}else{
  print("Less than Century")
}

> if(data$Score[1] >= 100){
+   print("Century")
+ }else{
+   print("Less than Century")
+ }
[1] "Century"
> |

```

6. Write an R program to determine whether Virat Kohli received **Man of the Match**.
- If Yes $\rightarrow$ "Award Received"
 - Else $\rightarrow$ "No Award"

```

    if(data$Man.of.the[1] == "Yes"){
      print("Award Received")
    }else{
      print("No Award")
    }
  }

~ ~
> if(data$Man.of.the[1] == "Yes"){
+   print("Award Received")
+ }else{
+   print("No Award")
+ }
[1] "No Award"
> |

```

7. Write an R program to check whether Virat Kohli was the **Captain**.

- If Yes → "Captain"
- Else → "Player"

```

if(data$Captain[1] == "Yes"){
  print("Captain")
}else{
  print("Player")
}

```

```

~ ~
> if(data$Captain[1] == "Yes"){
+   print("Captain")
+ }else{
+   print("Player")
+ }
[1] "Player"
~ ~

```

8. Write an R program to check the match result.

- If Won → "Successful Match"
- Else → "Match Not Won"

```

if(data$Result[1] == "Won"){
  print("Successful Match")
}else{
  print("Match Not Won")
}

~ ~
> if(data$Result[1] == "Won"){
+   print("Successful Match")
+ }else{
+   print("Match Not Won")
+ }
[1] "Match Not Won"
~ ~

```

C. Questions using if...else if...else (9–12)

9. Write an R program to classify the first score as:

- $\geq 150 \rightarrow$ "Very High Score"
- 100–149 $\rightarrow$ "Century"
- 50–99 $\rightarrow$ "Half Century"
- Otherwise $\rightarrow$ "Low Score"

```
if(data$Score[1] >= 150){
  print("Very High Score")
}else if(data$Score[1] >= 100){
  print("Century")
}else if(data$Score[1] >= 50){
  print("Half Century")
}else{
  print("Low Score")
}
> if(data$Score[1] >= 150){
+   print("Very High Score")
+ }else if(data$Score[1] >= 100){
+   print("Century")
+ }else if(data$Score[1] >= 50){
+   print("Half Century")
+ }else{
+   print("Low Score")
+ }
[1] "Century"
```

10. Write an R program to classify the **Strike Rate**:

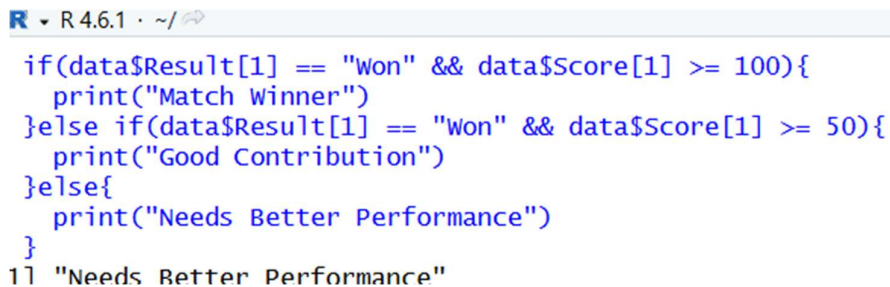
- $\geq 100 \rightarrow$ "Excellent"
- 80–99 $\rightarrow$ "Good"
- 60–79 $\rightarrow$ "Average"
- Below 60 $\rightarrow$ "Needs Improvement"

```
if(data$Strike.Rate[1] >= 100){
  print("Excellent")
}else if(data$Strike.Rate[1] >= 80){
  print("Good")
}else if(data$Strike.Rate[1] >= 60){
  print("Average")
}else{
  print("Needs Improvement")
}
```

11. Write an R program to classify the batting performance:

- Won + Century → "Match Winner"
- Won + Half Century → "Good Contribution"
- Lost → "Needs Better Performance"

```
if(data$Result[1] == "Won" && data$Score[1] >= 100){  
  print("Match Winner")  
}else if(data$Result[1] == "Won" && data$Score[1] >= 50){  
  print("Good Contribution")  
}else{  
  print("Needs Better Performance")  
}
```



```
R 4.6.1 · ~/   
if(data$Result[1] == "won" && data$Score[1] >= 100){  
  print("Match Winner")  
}else if(data$Result[1] == "won" && data$Score[1] >= 50){  
  print("Good Contribution")  
}else{  
  print("Needs Better Performance")  
}  
11 "Needs Better Performance"
```

12. Write an R program to classify the match format:

- Test → "Longest Format"
- ODI → "50 Overs"
- T20I → "20 Overs"
- Otherwise → "Other Format"

```
if(data$Format[1] == "Test"){  
  print("Longest Format")  
}else if(data$Format[1] == "ODI"){  
  print("50 Overs")  
}else if(data$Format[1] == "T20I"){  
  print("20 Overs")  
}else{  
  print("Other Format")  
}
```

```

> if(data$Format[1] == "Test"){
+   print("Longest Format")
+ }else if(data$Format[1] == "ODI"){
+   print("50 Overs")
+ }else if(data$Format[1] == "T20I"){
+   print("20 Overs")
+ }else{
+   print("Other Format")
+ }
[1] "Longest Format"

```

D. Questions using ifelse() (13–15)

13. Create a new column **Performance** using ifelse().

- Score $\geq 100 \rightarrow$ "Century"
- Otherwise $\rightarrow$ "Others"

```
data$Performance <- ifelse(data$Score >= 100, "Century", "Others")
```

```
print(data)
```

22	Test	No	Yes	Century
23	Test	Yes	Yes	Century
24	Test	No	Yes	Century
25	Test	No	Yes	Century
26	Test	Yes	Yes	Century
27	Test	No	Yes	Century
28	ODI	No	No	Century
29	ODI	Yes	No	Century
30	ODI	Yes	No	Century
31	ODI	Yes	No	Century
32	ODI	No	No	Century
33	ODI	No	No	Century
34	ODI	Yes	No	Century
35	ODI	Yes	No	Century
36	ODI	Yes	No	Century
37	ODI	Yes	No	Century
38	ODI	Yes	No	Century
39	ODI	Yes	No	Century
40	ODI	Yes	No	Century
41	ODI	Yes	Yes	Century
42	ODI	Yes	Yes	Century
43	ODI	No	No	Century
44	ODI	Yes	No	Century
45	ODI	No	No	Century
46	ODI	Yes	Yes	Century
47	ODI	Yes	No	Century
48	ODI	No	Yes	Century
49	ODI	Yes	No	Century
50	ODI	Yes	No	Century
51	ODI	No	No	Century
52	ODI	No	No	Century
53	ODI	Yes	No	Century
54	ODI	No	Yes	Century
55	ODI	Yes	Yes	Century
56	ODI	Yes	Yes	Century
57	ODI	No	Yes	Century
58	ODI	No	Yes	Century
59	ODI	No	Yes	Century
60	ODI	Yes	Yes	Century
61	ODI	Yes	Yes	Century

14. Create a new column **Winning Status** using ifelse().

- Result = "Won" → "Victory"
- Otherwise → "Not Victory"

```
data$Winning.Status <- ifelse(data$Result == "Won", "Victory", "Not Victory")
```

```
print(data)
```

26	Test	Yes	Yes	Century	Victory
27	Test	No	Yes	Century	Victory
28	ODI	No	No	Century	Victory
29	ODI	Yes	No	Century	Victory
30	ODI	Yes	No	Century	Victory
31	ODI	Yes	No	Century	Victory
32	ODI	No	No	Century	Victory
33	ODI	No	No	Century	Not Victory
34	ODI	Yes	No	Century	Victory
35	ODI	Yes	No	Century	Victory
36	ODI	Yes	No	Century	Victory
37	ODI	Yes	No	Century	Victory
38	ODI	Yes	No	Century	Victory
39	ODI	Yes	No	Century	Victory
40	ODI	Yes	No	Century	Victory
41	ODI	Yes	Yes	Century	Not Victory
42	ODI	Yes	Yes	Century	Victory
43	ODI	No	No	Century	Victory
44	ODI	Yes	No	Century	Victory
45	ODI	No	No	Century	Not Victory
46	ODI	Yes	Yes	Century	Victory
47	ODI	Yes	No	Century	Victory
48	ODI	No	Yes	Century	Victory
49	ODI	Yes	No	Century	Victory
50	ODI	Yes	No	Century	Victory
51	ODI	No	No	Century	Not Victory
52	ODI	No	No	Century	Not Victory
53	ODI	Yes	No	Century	Victory
54	ODI	No	Yes	Century	Victory
55	ODI	Yes	Yes	Century	Victory
56	ODI	Yes	Yes	Century	Victory
57	ODI	No	Yes	Century	Victory
58	ODI	No	Yes	Century	Not Victory

15. Create a new column **Captain Status** using ifelse().

- Captain = "Yes" → "Captain"
- Otherwise → "Player"

```
data$Captain.Status <- ifelse(data$Captain == "Yes", "Captain", "Player")
```

```
print(data)
```

	Man.of.the.Match	Captain	Performance	Winning.Status	Captain.Status
	No	No	Century	Not Victory	Player
	Yes	No	Century	Victory	Player
	No	No	Century	Not Victory	Player
	No	No	Century	Victory	Player
	No	No	Century	Not Victory	Player
	No	No	Century	Not Victory	Player
	No	Yes	Century	Not Victory	Captain
	No	Yes	Century	Not Victory	Captain
	No	No	Century	Not Victory	Player
0	No	Yes	Century	Not Victory	Captain
1	No	Yes	Century	Not Victory	Captain
2	No	Yes	Century	Victory	Captain
3	No	Yes	Century	Victory	Captain
4	Yes	Yes	Century	Victory	Captain
5	Yes	Yes	Century	Victory	Captain
6	Yes	Yes	Century	Victory	Captain
7	No	Yes	Century	Victory	Captain
8	No	Yes	Century	Not Victory	Captain
9	Yes	Yes	Century	Victory	Captain
0	Yes	Yes	Century	Not Victory	Captain
1	No	Yes	Century	Not Victory	Captain
2	No	Yes	Century	Not Victory	Captain
3	Yes	Yes	Century	Victory	Captain
4	No	Yes	Century	Victory	Captain
5	No	Yes	Century	Not Victory	Captain
6	Yes	Yes	Century	Victory	Captain
7	No	Yes	Century	Victory	Captain
8	No	No	Century	Victory	Player

E. Questions using for Loop (16–17)

16. Write an R program using a **for loop** to print all scores greater than or equal to **100**.

```
sc <- data$Score
for(i in sc){
  if(i >= 100){
    print(i)
  }
}
```

```

R ▾ R 4.6.1 · ~/
> sc <- data$Score
>
> for(i in sc){
+   if(i >= 100){
+     print(i)
+   }
+ }
[1] 116
[1] 103
[1] 103
[1] 107
[1] 119
[1] 105
[1] 115
[1] 141
[1] 169
[1] 147
[1] 103
[1] 200
[1] 211
[1] 167
[1] 235
[1] 204
[1] 103
[1] 104
[1] 213
[1] 243
[1] 153
[1] 149
[1] 103
[1] 139
[1] 123
[1] 254
[1] 136
[1] 107
[1] 102
[1] 118
[1] 105
[1] 100
[1] 107
[1] 112

```

17. Write an R program using a **for loop** to count how many matches Virat Kohli won.

```

r <- data$Result
count <- 0
for(i in r){
  if(i == "Won"){
    count <- count + 1
  }
}
print(count)

```

```

[1] 122
> r <- data$Result
> count <- 0
>
> for(i in r){
+   if(i == "Won"){
+     count <- count + 1
+   }
+ }
>
> print(count)
[1] 48

```

F. Questions using while Loop (18–19)

18. Write an R program using a **while loop** to display scores one by one until the first century is encountered.

```
i <- 1
while(i <= nrow(data)){
  print(data$Score[i])
  if(data$Score[i] >= 100){
    break
  }
  i <- i + 1
}
> while(i <= nrow(data)){
+   print(data$Score[i])
+   if(data$Score[i] >= 100){
+     break
+   }
+   i <- i + 1
+ }
[1] 116
```

19. Write an R program using a **while loop** to count the number of matches played in **Test** format.

```
i <- 1
count <- 0
while(i <= nrow(data)){
  if(data$Format[i] == "Test"){
    count <- count + 1
  }
  i <- i + 1
}
print(count)
>
> while(i <= nrow(data)){
+   if(data$Format[i] == "Test"){
+     count <- count + 1
+   }
+   i <- i + 1
+ }
>
> print(count)
```

G. Question using repeat Loop (20)

20. Write an R program using a **repeat loop** to print the names of all opponent teams (Against) one by one. Stop the loop when the opponent is **Australia** or when all records have been processed.

These questions progressively cover:

- **if** – 4 questions
- **if...else** – 4 questions
- **if...else if...else** – 4 questions
- **ifelse()** – 3 questions
- **for** – 2 questions
- **while** – 2 questions
- **repeat** – 1 question

They are suitable for beginners to intermediate R programming laboratory exercises using the provided Virat Kohli dataset.

```
> i <- 1
>
> repeat{
+   if(i > nrow(data)){
+     break
+   }
+   print(data$Against[i])
+   if(data$Against[i] == "Australia"){
+     break
+   }
+   i <- i + 1
+ }
[1] "Australia"
```

```
i <- 1
repeat{
  if(i > nrow(data)){
    break
  }
  print(data$Against[i])
  if(data$Against[i] == "Australia"){
    break
  }
  i <- i + 1
}
```